One Tensor Commitment for Tables of Every Height

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Abstract

We commit every table of a multi-table AIR proof in one tensor commitment of the Ligero and Brakedown kind: one Merkle tree per oracle whose leaf hashes every table's part of a codeword column, one fold vector, one set of t opened columns, and one sumcheck that reduces every table's evaluation claims to a shared point. Committed separately, T tables pay T fold vectors and T query sets. A table of 2^n rows sits on a grid of 2^c columns: a shorter table packs 2^{c-n} of its columns into one message row, a taller one keeps 2^{n-c} grid rows, and zero extension to a common height makes the pool one table, whose proximity error is the single-table error with $\Sigma + m - 1$ batched units for Σ slot roles and m masters. We prove that the commitment is evaluation binding in the random oracle model, with an extractor and the error stated in full, and that in zero-knowledge mode the verifier's view is simulatable given a zero-knowledge argument for affine and multiplicative relations over one-time-padded scalars, which Ligero instantiates. Merkle leaves without salts need $t + 1$ random support cells per row, one more than the query count. In Hekate's zero-knowledge mode, against separate commitments, proof size changes by -36 to -56% on ML-KEM-768, ML-DSA and AES and by -88% on RSA-2048.

1 Introduction

A proof over T AIR tables can open all of them with one fold vector and one set of queried columns. We commit every table of a proof in one Merkle tree per oracle and reduce every table's evaluation claims with one sumcheck.

The setting is a tensor polynomial commitment in the style of Ligero and Brakedown [1, 10] over the binary tower $\mathbb{F}_{2^{128}}$ [4], with a Reed–Solomon row code evaluated by the additive FFT [14] and claims on bit-packed columns reduced by ring-switching [8]. Each table carries its own ZeroCheck, and logUp buses [11] connect the tables. Every table then owes evaluation claims on its committed columns at points of its own height.

Committed separately, table i pays a Merkle root, a fold vector of $S + 2^{c_i}$ field elements for its own split $c_i \le n_i$, and t opened columns with their Merkle multiproof. For the seven tables of an ML-DSA-65 verification this layout produced a zero-knowledge proof of 2 959 KiB.

In the pool, a split c fixes one grid of 2^c columns for every table. A table of 2^n rows with $n \ge c$ keeps 2^{n-c} grid rows; a table with $n < c$ packs 2^{c-n} of its columns side by side into one message row, and the $c - n$ bits that select a column become variables of the evaluation sumcheck. One leaf hashes every table's digest of one codeword column. One sumcheck over $v = \max(c, \max_i n_i)$ variables carries every table, a table with fewer variables entering the remaining rounds through the factor $\mathrm{eq}(z, 0)$. After zero extension the pool is one table of 2^v rows whose slots vanish above their table's height (Lemma 7), and its proximity error is the single-table error on that shape (Section 4).

Four ordering rules carry the construction, each consumed by one step of the soundness proof (Remark 12): the slot challenge η follows every table's claims, every master carries its own power of the line challenge λ, a finished table is extended by $\mathrm{eq}(z, 0)$ and never by a constant, and a short table's block bits are sumcheck variables. The split minimizes an integer byte count (Section 5). Leaf hiding needs one support cell more than the query count (Section 6): at $S = t$ an unopened leaf of a tree without blind rows is a function of the witness, which a guess can test against the root.

Definitions 2 to 4 fix evaluation binding, honest-verifier zero knowledge and the outer argument. Theorem 6 is completeness. Theorem 10 proves evaluation binding in the random oracle model: every claim of every table is bound to the columns decoded from the pool's commitment, with error $\varepsilon_{\mathrm{prox}} + ((1+\rho)/2)^t$ for the commitment, terms without the factor W for the batching and the extraction term of Lemma 1; Lemma 8 derives $\varepsilon_{\mathrm{prox}}$ from the curve bound of [2] and the interleaving and tensor bounds of [7]. Theorem 11 carries the bound into zero-knowledge mode with the error of the outer argument added. Theorem 19 simulates the verifier's view without the witness given the outer argument's simulator.

In Hekate's zero-knowledge mode the pool changes proof size by -36 to -56% on ML-KEM-768, ML-DSA and AES and by -88% on RSA-2048 (Section 7).

Batching polynomials of one length under one commitment is the interleaved setting of Ligero and Brakedown, and the proximity bounds are those of [2] and [7]. Matrices of differing heights are committed under one Merkle tree in FRI stacks [19] and in S-two, whose whitepaper shows cross-domain correlated agreement for multi-table circle FRI [5], and claims on columns of differing heights are reduced inside the sumcheck by a jagged reduction [12, 9]. Beneath that reduction, SP1 Hypercube cuts the trace into columns of one fixed height and commits them as one interleaved batch in the manner of Ligero [17]; Binius64 lifts each shorter oracle to the common variable count by zero padding, the lift carrying the factor $\mathrm{eq}(0, \cdot)$ on the high variables, and batches the lifted oracles in one sumcheck and one BaseFold [18, §§5.3–5.4]. Here the heights are resolved in the layout: a tall table keeps its grid rows, a short one packs its columns into one message row, zero extension in place makes the pool one interleaved codeword, and one fold vector, one query set and one proximity term cover every table. New here are the layout for tables of different heights under one fold vector, its soundness and hiding proofs, the four rules with the attacks their violations admit, the split objective, the support requirement for unsalted leaves, and the measurements. The per-table ZeroCheck and logUp arguments are unchanged and outside this note. In zero-knowledge mode the verifier's checks on one-time-padded scalars are proved by an outer argument, a zero-knowledge argument for affine and multiplicative relations over committed scalars (Definition 4); its soundness error and its simulator enter the theorems as parameters. Ligero [1] is such an argument, and [13] gives the deployed instance and its rows for ring-switched claims.

2 Notation and the row code

Throughout, $\mathbb{F} = \mathbb{F}_{2^{128}}$, built as a tower of quadratic extensions [4], and $|\mathbb{F}| = 2^{128}$. For $a, x \in \mathbb{F}^n$ put $\mathrm{eq}(a, x) = \prod_j (a_j x_j + (1 + a_j)(1 + x_j))$. A table of 2^n rows commits each column f as the multilinear polynomial in n variables that agrees with f on $\{0,1\}^n$, and $\tilde{f}(a) = \sum_{x \in \{0,1\}^n} \mathrm{eq}(a, x) f(x)$ is its value at a.

Fix a split c and put $L = 2^c$. A message is a pair (ξ, d) with a support $\xi \in \mathbb{F}^S$, uniform in zero-knowledge mode, and data $d \in \mathbb{F}^L$. Its codeword is the polynomial $\sum_{k<S} \xi_k X_k + \sum_{k<L} d_k X_{S+k}$ evaluated on an $\mathbb{F}_2$-subspace of W points, where $X_0, X_1, \ldots$ is the novel

polynomial basis of [14] over a Cantor basis, $\deg X_k = k$. The code has length $W = 2L$ when $S \le L$ and the query term below reaches 100 bits at that length. At $S = 288$ and $t = 287$ the term reads 82 bits at $c = 10$ and 100.1 at $c = 11$: a grid with $c \le 10$ takes L support cells at $W = 4L$, and every deployed split has $c \ge 12$. The row code is Reed–Solomon of dimension $S + L$ and length W, hence MDS, with minimum distance $W - S - L + 1$.

A table of 2^n rows with $n \ge c$ stores a column f on $R = 2^{n-c}$ grid rows: grid row $g \in \{0,1\}^{n-c}$ holds $f(x, g)$ for $x \in \{0,1\}^c$. The columns of a table are grouped into slots, one slot per column here, and each pair of a slot and a grid row is one message with its own support. The encoded messages form a matrix of W columns, and the commitment is a Merkle tree over those columns.

The verifier batches a table's claims into at most two master claims. Each unit u of the table, a committed column or a ring-switched group of bit columns, carries a power $\eta^{\sigma(u)}$ of a slot challenge η, and a slot's coefficient $\kappa_{k,s}$ in master k is the sum of the powers of its units assigned to k. Master k is $M_k = \sum_s \kappa_{k,s} D_s$, D_s the multilinear polynomial of slot s, and a sumcheck reduces the table's batched claims to the values $\mu_k = M_k(r)$ at one point r. The prover sends a fold vector

$$q = \sum_k \lambda^k \sum_s \kappa_{k,s} \sum_g \mathrm{eq}(r_{\mathrm{hi}}, g)\, (\xi_{s,g}, d_{s,g}),$$

where r_{hi} are the last $n - c$ coordinates of r and r_{lo} the first c. The verifier checks the line $\langle d_q, \mathrm{eq}(r_{\mathrm{lo}}, \cdot)\rangle = \sum_k \lambda^k \mu_k$ on the data part d_q of q, draws t columns j, and checks that the encoding of q at j equals the same combination of the opened symbols, with their Merkle paths.

The single-table analysis bounds two terms. The query term is $-t \log_2((1+\rho)/2)$ bits with $\rho = (S + L)/W$ [10, App. B, Thm. 7]. The proximity error is $(\theta + U - 1)W/|\mathbb{F}|$, with $\theta = \log_2 R$ the depth of the tensor over grid rows [7] and $U - 1$ the degree of the batching curve over U units [2, Thm. 1.5]; Lemma 8 proves this form for the pool and Section 4 counts U.

Interleaved codes and proximity gaps. Write C for the row code and $e = \lfloor (W - S - L)/2 \rfloor$ for its unique decoding radius. For a matrix $\mathcal{U} \in \mathbb{F}^{N \times W}$ let $d^N(\mathcal{U}, C^N)$ be the least number of columns on which $\mathcal{U}$ differs from a matrix whose every row is a codeword; $\mathcal{U}$ is e-close if $d^N(\mathcal{U}, C^N) \le e$, and then that matrix is unique, each row of $\mathcal{U}$ lying within e of one codeword, and we call it the decoding of $\mathcal{U}$. Two bounds carry Section 4, both in this unique-decoding regime. The curve bound [2, Thm. 1.5]: for words $u_0, \dots, u_l \in \mathbb{F}^W$, if $\Pr_z[d(\sum_{i \le l} z^i u_i, C) \le e] > lW/|\mathbb{F}|$ over $z \in \mathbb{F}$, then the matrix with rows $u_0, \dots, u_l$ is e-close. The tensor bound: by [7, Thms. 1 and 2] every interleaving C^N has the proximity gap for affine lines of C, with false-witness bound W, and by [7, Thm. 3] applied to C^N it has tensor-style gaps with the same parameters: for matrices $\mathcal{U}_0, \dots, \mathcal{U}_{2^\vartheta - 1} \in \mathbb{F}^{N \times W}$, if $\Pr_r[d^N(\sum_g \mathrm{eq}(r, g)\, \mathcal{U}_g, C^N) \le e] > \vartheta W/|\mathbb{F}|$ over $r \in \mathbb{F}^\vartheta$, then the matrix that stacks the $\mathcal{U}_g$ is e-close.

Model. The protocol is public-coin and interactive, and the theorems are stated for it; its Fiat–Shamir compilation in the random oracle model is accounted as in [3, 10]. The prover makes at most Q_H queries to H.

Lemma 1 (Merkle extraction)**.** *Let H be a random oracle with λ-bit digests. There is an extractor that, from the queries of a prover that outputs a Merkle root, outputs a matrix of*

symbols such that every opening the prover later produces against that root and the verifier accepts agrees with the matrix, except with probability at most $Q_H^2/2^\lambda$ [3].

Throughout, the committed matrix of a tree is the output of Lemma 1 on its root; Section 4 extends it by zero rows to the matrix $\mathcal{P}$. The claims, units and points below are those of Section 3.

Definition 2 (evaluation binding)**.** The protocol is evaluation binding with error ε if for every prover the verifier accepts while one of the following holds with probability at most ε: (a) $\mathcal{P}$ is not e-close; (b) $\mathcal{P}$ is e-close with decoding $\mathcal{P}^*$ and a claim of some table differs from the value it names on $\mathcal{P}^*$, that is, for a whole unit $c_u \neq \tilde{f}^*(a_i)$ or $c'_u \neq \tilde{f}^*_+(a_i)$ and for a ring unit $c_{u,v} \neq \tilde{t}^*_v(a_i)$ or $c'_{u,v} \neq \tilde{t}^*_{v,+}(a_i)$, where f^* is the unit's column in $\mathcal{P}^*$ and t^*_v its v-th tower coordinate.

Definition 3 (honest-verifier zero knowledge)**.** The view of the verifier in zero-knowledge mode is the roots, the opened symbols with their paths, the fold vector q, and the pad commitment, the masked scalars and the messages of the outer argument. The protocol is honest-verifier zero-knowledge with advantage δ if a simulator with oracle access to H outputs, from the public inputs and the verifier's coins, a view that no distinguisher making Q_H queries to H tells from the honest view with advantage above δ, for every witness.

Definition 4 (outer argument)**.** An outer argument is a zero-knowledge argument for a statement of affine rows $\sum_j a_j v_j = b$ and multiplicative rows $v_i v_j = v_k$ with public coefficients over a vector v of scalars. The prover commits to a pad p before the first challenge, sends the masked scalars $v+p$ in place of v, and proves the rows. It is ε_{out}-sound if for every prover the verifier accepts while a row fails on $(v+p)-p^*$, p^* the pad its commitment binds (for a Merkle tree over the encoded pad, the decoding of the rows of Lemma 1), with probability at most ε_{out}. It is δ_{out}-zero-knowledge if a simulator with the verifier's coins and the statement, without v, outputs a pad commitment, masked scalars and messages that no distinguisher making Q_H queries to H tells from the honest ones with advantage above δ_{out}. Ligero [1] is one, and [13] is the deployed instance.

3 The pool

The pool is every table of a proof in a fixed order, T tables with 2^{n_i} rows each. The split c is derived in Section 5; put $v = \max(c, \max_i n_i)$ and $v_i = \max(c, n_i)$. Table i uses $R_i = 2^{\max(n_i - c, 0)}$ grid rows and packs $P_i = 2^{\max(c - n_i, 0)}$ columns per slot.

A table with $n_i \geq c$ is laid out as in Section 2. A table with $n_i < c$ has one grid row, and its slot holds P_i columns side by side: column (s, b) for $b \in \{0,1\}^{c-n_i}$ occupies positions $b \cdot 2^{n_i} + x$, and $D_s(x, b)$ is a multilinear polynomial in c variables. A class of columns fills $\lceil \text{count}/P_i \rceil$ slots, the last closed by zero blocks. In zero-knowledge mode table i adds KP_i whole blind columns, $K = 1$ in the deployed system, and its ring-switched units a blind of their own [6]. The slot roles of every table are numbered by one index σ running over the pool in table order, Σ in all, and the pool has m masters, one or two per table.

Every message of every table is encoded at the same width W. Leaf j of the trace tree is

$$H(\chi_{0,j} \,\|\, \cdots \,\|\, \chi_{T-1,j} \,\|\, \texttt{0x02}), \qquad \chi_{i,j} = H(\text{part}_{i,j} \,\|\, \texttt{0x00}),$$

where $\text{part}_{i,j}$ is the column-j symbols of table i over its slots and grid rows, and an inner node hashes its two children and `0x01`. The verifier hashes $T+1$ times per opened column. The logUp helper columns h of the bus-carrying tables form slots in a second tree of the same width, committed after the bus challenges; a pool with no bus has none. The fold vector and the queries cover both trees.

Protocol. The prover sends the roots, the claims, the round polynomials, the masters, the fold vector and the openings; the verifier draws each challenge after the message it follows and checks as stated below. The transcript runs in this order.

1. The main table's dimensions, then each chiplet's in pool order, then the trace root; in zero-knowledge mode, the root of the one-time pad.
2. The logUp challenges γ, β; every lookup bus's maximal height, then each bus's point; then the root of the h tree.
3. For each table in pool order: its claimed bus sums, its ZeroCheck challenge and rounds, its h evaluations and its column claims.
4. The slot challenge η; $\max_i(c - n_i)$ block challenges, of which table i reads the first $c - n_i$ as ρ_i; the ring-switching challenge when a table carries ring units; the eval sumcheck; the masters $\tilde{\mu}_k$ in pool order; λ when $m > 1$; the fold vector q; the t query columns; both openings.

For table i split $y \in \mathbb{F}^v$ as $y = (x, b, z)$ with $x \in \mathbb{F}^{n_i}$, $b \in \mathbb{F}^{v_i - n_i}$ and $z \in \mathbb{F}^{v - v_i}$; a tall table has no b, and a table with $v_i = v$ has no z. The eval sumcheck runs over v variables, low variable first, on

$$F(y) = \sum_i \operatorname{eq}(z, 0) \operatorname{eq}(\rho_i, b) \sum_{k \in K_i} \omega_k(x)\, M_k(x, b), \tag{1}$$

where K_i are the masters of table i and ω_k are the public multilinear weights defined below. Its initial claim is $\sum_i \tau_i$, the batched value of every table's claims, formed as follows.

Table i claims its columns at one point $a_i \in \mathbb{F}^{n_i}$, the point its ZeroCheck ended at, and at the next row: a column f claims the pair $(\tilde{f}(a_i), \tilde{f}_+(a_i))$, $f_+(x) = f(x+1)$ the column shifted by one row. A unit is a whole column f, or a group of $J \le 128$ bit columns $t_0, \dots, t_{J-1}$ that one column packs as $f = \sum_v \pi_v t_v$ over the tower basis (π_v) of $\mathbb{F}$ over $\mathbb{F}_2$; a ring unit claims the $2J$ values $\tilde{t}_v(a_i)$ and $\tilde{t}_{v,+}(a_i)$. Every unit u has a slot $s(u)$, a block $b(u)$ of that slot, a slot role $\sigma(u)$ and a master $k(u)$; the units of a slot fall into roles, ring roles and whole roles, each role holding at most one unit per block of the slot, and $\kappa_{k,s}$ sums η^σ over the roles σ of slot s in master k. Blocks of a slot that hold no unit are zero columns. Once every table has claimed, the verifier draws η, the block challenges ρ_i and, when ring units exist, $r'' \in \mathbb{F}^\nu$ with $\nu = \log_2 128 = 7$, and gives unit u the value

$$y_u = c_u + \eta^\Sigma c'_u \quad \text{or} \quad y_u = \sum_{v<J} \pi_v \big(\phi_{r''}(c_{u,v}) + \eta^\Sigma \phi_{r''}(c'_{u,v})\big),$$

c and c' the claims at a_i and at the next row, where $\phi_{r''}(x) = \sum_{w \in \{0,1\}^\nu} \operatorname{eq}(r'', w) \operatorname{bit}_w(x)$ batches the tower coordinates of $x \in \mathbb{F}$ and is $\mathbb{F}_2$-linear. Then

$$\tau_i = \sum_{u \in i} \eta^{\sigma(u)} \operatorname{eq}(\rho_i, b(u))\, y_u.$$

The weight of the whole master of table i is $\omega_k = \operatorname{eq}(a_i, \cdot) + \eta^\Sigma K(a_i, \cdot)$, where $K(a_i, \cdot)$ is the multilinear weight with $\sum_x K(a_i, x) f(x) = \tilde{f}_+(a_i)$ for every column f; the weight of the ring master is $\omega_k = \Phi_{r''}(\operatorname{eq}(a_i, \cdot)) + \eta^\Sigma \Phi_{r''}(K(a_i, \cdot))$, where $\Phi_{r''}(w)$ is the multilinear extension of $x \mapsto \phi_{r''}(w(x))$ on $\{0,1\}^{n_i}$, evaluated as in [13]. Each round polynomial has degree 2. At the final point $r = (r_x, r_b, r_z)$, read per table, the verifier checks

$$F(r) = \sum_i \operatorname{eq}(r_z, 0) \operatorname{eq}(\rho_i, r_b) \sum_{k \in K_i} \omega_k(r_x)\, \mu_k \tag{2}$$

against the masters the prover then sends, $\tilde{\mu}_k = \mathrm{eq}(r_z, 0)\,\mu_k$ for the master k of table i with μ_k the claimed value of $M_k(r_x, r_b)$. Let $r_{\mathrm{lo}}, r_{\mathrm{hi}}$ be the first c and the last $v - c$ coordinates of r. The fold vector is

$$q = \sum_s \Big(\sum_k \lambda^k \kappa_{k,s}\Big) \sum_{g < R_{i(s)}} \mathrm{eq}(r_{\mathrm{hi}}, g)\,(\xi_{s,g}, d_{s,g}), \tag{3}$$

with g read as a point of $\{0,1\}^{v-c}$, and the verifier checks the line

$$\langle d_q, \mathrm{eq}(r_{\mathrm{lo}}, \cdot)\rangle = \sum_k \lambda^k \tilde{\mu}_k \tag{4}$$

and, at each of the t drawn columns j, that the encoding of q equals the same combination (3) of the opened symbols of both trees.

Lemma 5. *For an honest prover,* (4) *holds.*

Proof. It suffices to show $\sum_{g<R_i} \mathrm{eq}(r_{\mathrm{hi}}, g)\langle d_{s,g}, \mathrm{eq}(r_{\mathrm{lo}}, \cdot)\rangle = \mathrm{eq}(r_z, 0) D_s(r_x, r_b)$ for each slot s of table i. If $n_i \geq c$, then $r_{\mathrm{lo}} = r_x$ restricted to its first c coordinates, and for $g < R_i$ the factor $\mathrm{eq}(r_{\mathrm{hi}}, g)$ splits as $\mathrm{eq}(r', g)\,\mathrm{eq}(r_z, 0)$, r' the coordinates $c, \ldots, n_i - 1$ of r; the sum over g and the inner product together evaluate D_s at r_x. If $n_i < c$, then $R_i = 1$, $r_{\mathrm{lo}} = (r_x, r_b)$ and $\mathrm{eq}(r_{\mathrm{hi}}, 0) = \mathrm{eq}(r_z, 0)$, and the inner product is $D_s(r_x, r_b)$ by the layout of packed columns. Summing over slots with the coefficients of (3) gives $\sum_k \lambda^k \tilde{\mu}_k$. □

Theorem 6 (completeness)**.** *For an honest prover the verifier accepts.*

Proof. Write $\langle \omega, f\rangle = \sum_x \omega(x) f(x)$ over the cube of the table. For a whole unit, $y_u = \tilde{f}(a_i) + \eta^{\Sigma} \tilde{f}_+(a_i) = \langle \omega_k, f\rangle$ by the definition of K. For a ring unit, $f(x) = \sum_v \pi_v t_v(x)$ with $t_v(x) \in \mathbb{F}_2$, and $\phi_{r''}$ is $\mathbb{F}_2$-linear, whence

$$\sum_x \phi_{r''}(\mathrm{eq}(a_i, x))\, f(x) = \sum_v \pi_v\, \phi_{r''}\Big(\sum_x \mathrm{eq}(a_i, x)\, t_v(x)\Big) = \sum_v \pi_v\, \phi_{r''}(\tilde{t}_v(a_i)),$$

and likewise with $K(a_i, \cdot)$ and $t_{v,+}$; again $y_u = \langle \omega_k, f\rangle$. A unit of table i in slot s at block b has $f = D_s(\cdot, b)$, a tall table having one block. Summing (1) over $\{0,1\}^v$, $\sum_z \mathrm{eq}(z, 0) = 1$, and since $\kappa_{k,s}$ sums η^{σ} over the roles σ of slot s in master k and a role's units run over the blocks of its slot, the other blocks holding zero columns,

$$\sum_y F(y) = \sum_i \sum_{k \in K_i} \sum_s \kappa_{k,s} \sum_b \mathrm{eq}(\rho_i, b)\langle \omega_k, D_s(\cdot, b)\rangle = \sum_i \sum_{u \in i} \eta^{\sigma(u)}\, \mathrm{eq}(\rho_i, b(u))\, y_u = \sum_i \tau_i.$$

The round polynomials of F have degree at most 2 in each variable, and $F(r)$ is (2) with $\mu_k = M_k(r_x, r_b)$. The line is Lemma 5. The encoding is linear, and the encoding of q at a column is the combination (3) of the rows' symbols there. □

4 Soundness of the pool

To the verifier the pool is one table, and its proximity error is the single-table error with $U = \Sigma + m - 1$. Let $\mathcal{P}$ be the matrix with one row for each pair (s, g) of a slot of the pool and a point $g \in \{0,1\}^{v-c}$, equal to the committed row of slot s on grid row g, its W symbols, when $g < R_{i(s)}$ and to zero otherwise.

Lemma 7. *On the line and the column checks, the verifier of the pool runs the verifier of one table on $\mathcal{P}$, with grid depth $\theta = v - c$ and fold weights $\big(\sum_k \lambda^k \kappa_{k,s}\big)\mathrm{eq}(r_{\mathrm{hi}}, g)$, and every row of $\mathcal{P}$ outside the commitment is zero in every column.*

Proof. A zero row is a codeword and adds nothing to any column. Summing the opened symbols over the committed rows, as the column check does, therefore equals summing over every row of $\mathcal{P}$. For an honest prover the rows of $\mathcal{P}$ encode the messages of (3), the encoding of q is the fold of $\mathcal{P}$ by the stated weights, and by Lemma 5 the line (4) is the line of that table at r with the zero-extended masters $\tilde{\mu}_k$. □

Write $q^\circ \in \mathbb{F}^W$ for the fold of $\mathcal{P}$ itself, whose entry j is the combination (3) of the committed symbols at column j; the column check compares the encoding of q with q° at the drawn columns. Each row of $\mathcal{P}$ carries the weight $w_s \operatorname{eq}(r_{\mathrm{hi}}, g)$ with $w_s = \sum_k \lambda^k \kappa_{k,s}$.

Lemma 8 (proximity)**.** *If $\mathcal{P}$ is not e-close, then q° is e-close with probability at most*

$$\varepsilon_{\mathrm{prox}} = (\theta + \Sigma + m - 2)\, W/|\mathbb{F}| \tag{5}$$

over η, r_{hi} and λ.

Proof. Fold in three stages: $\mathcal{P}' \in \mathbb{F}^{\Sigma \times W}$ with rows $\mathcal{P}'_s = \sum_g \operatorname{eq}(r_{\mathrm{hi}}, g)\, \mathcal{P}_{s,g}$, the tensor fold of the 2^θ matrices $(\mathcal{P}_{s,g})_s$; then $c_k = \sum_s \kappa_{k,s} \mathcal{P}'_s$ for each master k; then $q^\circ = \sum_k \lambda^k c_k$. Let E_1 be the event that $\mathcal{P}'$ is e-close, E_2 that the matrix with rows $c_0, \ldots, c_{m-1}$ is, and E_3 that q° is. The three challenges are uniform and independent whatever the prover sends between them, and $\mathcal{P}$ precedes them all.

Since $\mathcal{P}$ is not e-close, the tensor bound for C^Σ with $\vartheta = \theta$ gives $\Pr[E_1] \le \theta W/|\mathbb{F}|$. Fix r_{hi} outside E_1. If the c_k agree with codewords v_k outside e common columns, then $\sum_k c_k$ agrees with the codeword $\sum_k v_k$ there: E_2 implies that $\sum_k c_k$ is e-close; and $\sum_k c_k = \sum_s (\sum_k \kappa_{k,s}) \mathcal{P}'_s = \sum_{\sigma < \Sigma} \eta^\sigma \mathcal{P}'_{s(\sigma)}$ is a curve of degree $\Sigma - 1$ in η whose coefficient words are rows of $\mathcal{P}'$, every row occurring since every slot holds a unit. By the curve bound, $\Pr_\eta[E_2] \le (\Sigma - 1)W/|\mathbb{F}|$, as a larger probability would make $\mathcal{P}'$ e-close. Fix η outside E_2. Then q° is a curve of degree $m-1$ in λ with coefficient words c_k, and the curve bound gives $\Pr_\lambda[E_3] \le (m-1)W/|\mathbb{F}|$. The union bound over the three stages is (5). □

Lemma 9 (column test)**.** *Let the t columns be drawn uniformly and independently. If q° is not e-close, every column check passes with probability at most $((1+\rho)/2)^t$. If $\mathcal{P}$ is e-close with decoding $\mathcal{P}^*$ and q differs from the fold q^* of the messages of $\mathcal{P}^*$ by the weights of* (3)*, every column check passes with probability at most $((1+\rho)/2)^t$.*

Proof. The check at column j reads $\operatorname{Enc}(q)_j = q^\circ_j$. In the first case $\operatorname{Enc}(q)$ is a codeword, from which q° differs on more than e columns, and one check passes with probability at most $1 - (e+1)/W \le (1+\rho)/2$. In the second case let A be the set of at most e columns on which $\mathcal{P}$ and $\mathcal{P}^*$ differ. Outside A the symbols of $\mathcal{P}$ are those of $\mathcal{P}^*$, and the combination of codewords is the codeword of the combined messages, whence $q^\circ_j = \operatorname{Enc}(q^*)_j$ for $j \notin A$. The distinct codewords $\operatorname{Enc}(q)$ and $\operatorname{Enc}(q^*)$ agree on at most $S + L - 1$ columns, and a check passes only on A or on those columns, $e + S + L - 1 \le (1+\rho)W/2$ in all. The t draws are independent. □

Theorem 10 (soundness)**.** *Let $\bar{p} = \max_i \max(c - n_i, 0)$, and let $\nu = 7$ when some table carries ring units and $\nu = 0$ otherwise. In base mode the protocol is evaluation binding with error $\varepsilon_{\mathrm{pool}}$ plus the term of Lemma 1 per tree, where*

$$\varepsilon_{\mathrm{pool}} = \varepsilon_{\mathrm{prox}} + \Big(\frac{1+\rho}{2}\Big)^t + \frac{(m-1) + 2v + (2\Sigma - 1 + \bar{p} + \nu)}{|\mathbb{F}|}.$$

Proof. The openings agree with $\mathcal{P}$ except for the term of Lemma 1, charged once per tree. Six steps follow, each charged once, over the events of Definition 2.

1. Let $\mathcal{P}$ not be e-close. By Lemma 8, q° is e-close with probability at most $\varepsilon_{\text{prox}}$, and otherwise Lemma 9 rejects except with probability $((1+\rho)/2)^t$. Henceforth $\mathcal{P}$ is e-close with decoding $\mathcal{P}^*$; its rows outside the commitment are zero (Lemma 7), and its messages define the decoded slot polynomials D^*_s, masters $M^*_k = \sum_s \kappa_{k,s} D^*_s$, columns f^*_u and coordinates $t^*_{u,v}$.

2. If $q \neq q^*$, Lemma 9 rejects except with probability $((1+\rho)/2)^t$. The matrix $\mathcal{P}$ is fixed; steps 1 and 2 are exclusive, and the query term is charged once. Henceforth $q = q^*$.

3. Put $\tilde{\mu}^*_k = \text{eq}(r_z, 0)\, M^*_k(r_x, r_b)$. The identity in the proof of Lemma 5 holds for every message matrix; with $q = q^*$ the left side of the line (4) is $\sum_k \lambda^k \tilde{\mu}^*_k$. The masters $\tilde{\mu}_k$ precede λ; if some $\tilde{\mu}_k \neq \tilde{\mu}^*_k$, the line holds only at a root of the nonzero polynomial $\sum_k \lambda^k(\tilde{\mu}_k - \tilde{\mu}^*_k)$ of degree below m, probability at most $(m-1)/|\mathbb{F}|$. Henceforth $\tilde{\mu}_k = \tilde{\mu}^*_k$ for every k.

4. Let F^* be (1) with the masters M^*_k. With $\tilde{\mu}_k = \tilde{\mu}^*_k$ the final check (2) reads $F^*(r)$, and the verifier holds every round polynomial to degree 2. If $\sum_i \tau_i \neq \sum_{y \in \{0,1\}^v} F^*(y)$, the sumcheck accepts with probability at most $2v/|\mathbb{F}|$ [15]. Henceforth $\sum_i \tau_i = \sum_y F^*(y)$.

5. Summing F^* over $\{0,1\}^v$ as in the proof of Theorem 6 gives $\sum_y F^*(y) = \sum_i \tau^*_i + X$. Here $\tau^*_i = \sum_{u \in i} \eta^{\sigma(u)} \text{eq}(\rho_i, b(u))\, y^*_u$ with y^*_u the value y_u takes on the true claims of f^*_u, the identity of a ring unit holding for the coordinates $t^*_{u,v}$ of any column of $\mathcal{P}^*$; and

$$X = \sum_\sigma \eta^\sigma \sum_{b \notin U_\sigma} \text{eq}(\rho_{i(\sigma)}, b)\, \langle \omega_{k(\sigma)}, D^*_{s(\sigma)}(\cdot, b) \rangle,$$

U_σ the blocks that hold a unit of role σ, collects what $\mathcal{P}^*$ carries in the blocks an honest prover leaves zero.

6. The claims precede η, the ρ_i and r''. The difference $\sum_i (\tau_i - \tau^*_i) - X$ is a polynomial in these challenges. Sorted by the monomials η^σ and $\eta^{\Sigma+\sigma}$ and then by the block, its terms are $\text{eq}(\rho_i, b)\, \Delta^0_u$ and $\text{eq}(\rho_i, b)\, \Delta^1_u$ at a block b holding the unit u of role σ, and $-\text{eq}(\rho_i, b)$ times the η-free part and the η^Σ part of $\langle \omega_{k(\sigma)}, D^*_{s(\sigma)}(\cdot, b) \rangle$ at a block without one, where for a whole unit $\Delta^0_u = c_u - \tilde{f}^*_u(a_i)$ and $\Delta^1_u = c'_u - \tilde{f}^*_{u,+}(a_i)$, and for a ring unit $\Delta^0_u = \sum_v \pi_v\, \phi_{r''}(c_{u,v} - \tilde{t}^*_{u,v}(a_i))$ and Δ^1_u likewise with the next-row claims. A block holds at most one unit of a role, the $\text{eq}(\rho_i, b)$ for distinct b are linearly independent, $\phi_{r''}(x) = \sum_w \text{eq}(r'', w)\, \text{bit}_w(x)$ is the zero polynomial in r'' only at $x = 0$, and the π_v are a basis. Hence the difference is the zero polynomial only when every Δ vanishes, that is, when every claim is true, and otherwise it has degree at most $2\Sigma - 1$ in η, at most $\bar{p}$ in the ρ_i and at most ν in r'', and vanishes at the drawn challenges with probability at most $(2\Sigma - 1 + \bar{p} + \nu)/|\mathbb{F}|$. □

Theorem 11 (zero-knowledge mode)**.** *Let the outer argument be ε_{out}-sound. In zero-knowledge mode, where the claims, the masters and the round values are one-time-padded and the checks of steps 3, 4 and 6 are rows of its statement, the protocol is evaluation binding, with the claims of Definition 2 read as plaintexts, with error $\varepsilon_{\text{pool}} + \varepsilon_{\text{out}}$ plus the term of Lemma 1 per tree.*

Proof. The pad is committed before the first challenge, and a masked scalar fixes its plaintext, the masked value less the extracted pad, when it is sent. The verifier accepts only if the outer argument accepts, hence, except with probability ε_{out}, the plaintexts satisfy the line, every round check and the final check of the sumcheck, and the batching of the claims into $\sum_i \tau_i$. The column checks and the encoding of q run in the clear. The six steps of the proof of Theorem 10 then apply to the plaintexts, each precedence they use holding because the pad root precedes every challenge. For a ring unit $\phi_{r''}$ is $\mathbb{F}_2$-linear and

not $\mathbb{F}$-linear; the rows evaluate it as a linearized polynomial through its Frobenius chain of multiplicative rows [13]. □

Remark 12 (the four rules). Each ordering rule of the protocol is consumed by one step of the proof, and its violation admits a forgery or an inconsistency, exhibited in Appendix A. The slot challenge η follows the claims of every table: step 6 needs the claim polynomial fixed before η, and a prover that knows η first moves two claims by δ_1, δ_2 with $\delta_1\eta^{a_1}+\delta_2\eta^{a_2} = 0$, leaving τ_i and every later check unchanged. Every master carries its own power λ^k: step 3 separates the masters on the line, and under one pair $(1, \lambda)$ for the pool a claim error routed through the sumcheck into a whole master of table i and one of table i' cancels on the line in characteristic 2. A table with $v_i < v$ enters the rounds beyond v_i through $\mathrm{eq}(z, 0)$: step 5 uses $\sum_z \mathrm{eq}(z, 0) = 1$, and a constant a contributes $\sum_{z\in\{0,1\}^{v-v_i}} a = 0$ to the initial claim, dropping the table's claims from the check. The block bits of a short table are sumcheck variables: step 3 applies the identity of Lemma 5 to the decoded data, and a master opened at its own point (r_x, ρ_i) is another linear form on its slot than the one (4) applies to q, failing the line for an honest prover. The root of the h tree precedes the first ZeroCheck challenge of every table, the rule a single table already obeys, held across the pool.

Corollary 13 (extraction)**.** *With H a random oracle, the extractor that reads the leaves of both trees from the prover's oracle queries and decodes every row of $\mathcal{P}$ to radius e, by Berlekamp–Welch on the evaluation points of the row code, outputs columns whose evaluations at a_i and at the next row, and those of their tower coordinates, are the claims of every accepting transcript, except with probability $\varepsilon_{\mathrm{pool}}$ plus the term of Lemma 1 per tree.*

On the workloads of Table 1 the terms of $\varepsilon_{\mathrm{pool}}$ without the factor W sum to less than 2^{-118}, and the query term is unchanged from a single table: one code, one width W, t columns. The per-table ZeroCheck and logUp terms do not change.

Both parties compute the shape from absorbed data in integer arithmetic and reject one whose query or proximity term falls below 100 bits; the verifier holds the logUp term, 106 to 128 bits on the workloads below, to the same floor. Table 1 lists the shape of each workload of Section 7 in zero-knowledge mode. The proximity term is the smaller on every row, 102 to 108 bits; the query term is 109 to 118.

5 The split

The split c minimizes an integer estimate of the bytes that depend on it:

$$\mathrm{cost}(c) = 17 \cdot 2^c + t \sum_i 2^{\max(n_i - c, 0)}\, \ell_i(P_i) + 32K \sum_i (P_i - 1). \tag{6}$$

The first term is the data part of q, priced at 17 bytes per element, a one-byte bincode varint tag and 16 bytes; its support part, S elements from $c = 11$, is not priced. The second is the opened symbols: $\ell_i(P)$ is the byte count of table i's symbols on one grid row of one column, both trees, at P columns per slot. The third is the claims on blind columns: each of the KP_i blind columns of table i sends two masked claims of 16 bytes, and the term counts the $K(P_i - 1)$ blinds beyond those of an unpacked table. In base mode $K = 0$.

The search runs over $1 \le c \le \max_i(n_i + p_i)$, p_i the packing depth of table i's widest class, and admits c when it leaves room for the support, $2^c \ge S$, and when the shape of Section 4 clears 100 bits. Ties go to the narrower split. The inputs are the absorbed heights, the program and the configuration; nothing in the proof sets c.

workload	T	v	c	$\log_2 W$	U	query	proximity
ML-KEM-768 sender	9	13	13	14	229	114	106
ML-KEM-768 receiver	8	15	14	15	206	116	105
ML-DSA-44	7	15	14	15	226	116	105
ML-DSA-65	7	15	14	15	272	116	104
ML-DSA-87	7	16	15	16	239	117	104
AES-128	3	19	16	17	194	118	103
AES-256	3	19	16	17	234	118	103
RSA-2048	3	15	15	16	143	117	104
SHA-256, 2^{11} rows	1	12	12	13	91	109	108
SHA-256, 2^{21} rows	1	21	17	18	219	118	102
Keccak-f, 2^{15} rows	1	15	14	15	84	116	106
Keccak-f, 2^{20} rows	1	20	16	17	84	118	104
Fibonacci, 2^{20} rows	1	20	15	16	12	117	108
Fibonacci, 2^{24} rows	1	24	17	18	12	118	105
Fibonacci, 2^{26} rows	1	26	18	19	12	118	104

Table 1: Shape and security bits of the pool in zero-knowledge mode, as the verifier logs them, $t = 287$, $S = 288$. $U = \Sigma + m - 1$; the proximity column is $128 - \log_2 W - \lceil \log_2(v - c + U - 1) \rceil$.

Not counted: the Merkle multiproof, at most t hashes per tree per unit of c, and the eval sumcheck, $2v$ elements; the masters and the remaining claims do not depend on c. With both added, the minimizer moves on no workload of Table 1 in either mode, and under that count the next split costs at least 18.6 KiB more.

The third term matters for small tables: a table of 2^4 rows in a pool at $c = 16$ packs $P = 2^{12}$, and its 4 096 blind columns send 128 KiB of claims and 8 192 masked scalars to the outer argument. Every table of Table 1 has at least 2^9 rows, and the term stays below 4 KiB per table there.

6 Leaves and the support

Unsalted leaves need one support cell more than the query count. The random support is the device of zero-knowledge Ligero [1]; the leaf fixes its size. In zero-knowledge mode the verifier sees the fold vector q, data and support parts, the masked masters, and the opened symbols of at most t distinct columns. Every other column reaches it only through its Merkle leaf, a hash with no salt. If an unopened column is a function of the witness and the view, a guess of the witness can be tested against the root.

Lemma 14. *Let $S = t$. For every message (ξ, d) and every column j, the symbol at j is an affine function of d and the symbols at the t opened columns.*

Proof. The symbol at a point α is $\sum_{k<S} \xi_k X_k(\alpha) + \sum_{k<L} d_k X_{S+k}(\alpha)$. Since $\deg X_k = k$, the X_k with $k < S$ span the polynomials of degree below S, and a nonzero one has fewer than S roots. The $S \times t$ matrix $(X_k(\alpha_j))$ over the opened points is therefore invertible at $S = t$, and ξ is an affine function of d and the opened symbols. Substituting it gives the symbol at j. □

At $S = t$, then, a tree whose rows carry no uniform data cell has every unopened leaf fixed by the witness and the view. The h tree is such a tree: it carries no blind. The trace tree of a pool with one table, without ring units, on one grid row is another: there q fixes

the data of the one blind slot as q's data part minus the witness terms, and Lemma 14 fixes the rest.

Lemma 15. *Let $S \geq t+1$, and let at least two messages carry nonzero weight in q. Given the view and every data cell, the symbol of any message at any unopened column is uniform.*

Proof. With the data fixed, the view is linear in the supports ξ_r. Put $x_j = (X_k(\alpha_j))_{k<S}$ and let A be the span of $x_{j'}$ over the opened columns j'. Polynomials of degree below $S \geq t+1$ interpolate any $t+1$ values, whence $x_j \notin A$ for an unopened j. The view determines $\langle \xi_r, a\rangle$ for every r and $a \in A$, and $\sum_r w_r \xi_r$, w the weights of q. Were $\langle \xi_r, x_j\rangle$ determined as well, it would equal $\sum_{r'} \langle \xi_{r'}, a_{r'}\rangle + \langle \sum_{r'} w_{r'} \xi_{r'}, u\rangle$ identically, every $a_{r'} \in A$. A message $r' \neq r$ with $w_{r'} \neq 0$ forces $u \in A$, and then $x_j = a_r + w_r u \in A$, a contradiction. □

In the pool every table has a data slot and a blind slot, and each weight of (3) is a nonzero polynomial of degree below $\Sigma + m + \theta$ in the challenges: the hypothesis fails with probability at most $2(\Sigma + m + \theta)/|\mathbb{F}|$.

Theorem 19 assembles the lemmas into a simulator. Index the committed rows of both trees by r, $N \geq 2$ in all, each with a support ξ_r uniform in $\mathbb{F}^S$ and data d_r, the data of a blind row uniform too. The verifier opens $t' \leq t$ distinct columns $j_1, \ldots, j_{t'}$; with x_j as in Lemma 15 and $y_j = (X_{S+k}(\alpha_j))_{k<L}$, the symbol of row r at column j is $\langle \xi_r, x_j\rangle + \langle d_r, y_j\rangle$, $o_{r,l}$ denotes it at j_l, and w_r is the weight of row r in (3).

Lemma 16 (openings)**.** *For $t' \leq S$ the opened symbols of a row are uniform in $\mathbb{F}^{t'}$ and independent of its data, and the rows are independent.*

Proof. The $x_{j_1}, \ldots, x_{j_{t'}}$ are linearly independent, as in the proof of Lemma 15, and $\xi \mapsto (\langle \xi, x_{j_l}\rangle)_l$ is onto $\mathbb{F}^{t'}$: a uniform ξ_r makes the opened symbols uniform whatever d_r. □

Lemma 17 (the fold)**.** *Let $S \geq t'+1$, let $\bar{x}_1, \ldots, \bar{x}_{S-t'} \in \mathbb{F}^S$ complete $x_{j_1}, \ldots, x_{j_{t'}}$ to a basis, and let $\eta\lambda \neq 0$. Given the opened symbols of every row, the data part d_q of q is uniform in $\mathbb{F}^L$, the coordinates $\langle \xi_q, \bar{x}_m\rangle$ of its support part are uniform in $\mathbb{F}^{S-t'}$, the two are independent of each other and of the openings, and the remaining coordinates $\langle \xi_q, x_{j_l}\rangle = \sum_r w_r o_{r,l} - \langle d_q, y_{j_l}\rangle$ are determined by the openings and d_q.*

Proof. The last claim is the column check at j_l, which the honest prover satisfies. Put $\zeta_r = (\langle \xi_r, \bar{x}_m\rangle)_m \in \mathbb{F}^{S-t'}$; in the basis $x_{j_1}, \ldots, x_{j_{t'}}, \bar{x}_1, \ldots, \bar{x}_{S-t'}$ the coordinates of a uniform ξ_r are independent and uniform; the ζ_r are therefore uniform and independent of the openings, and $(\langle \xi_q, \bar{x}_m\rangle)_m = \sum_r w_r \zeta_r$ is uniform because some $w_r \neq 0$. The blind slot of a table with $v_i = v$ has weight $\lambda^k \eta^\sigma \neq 0$ in (3), and its data $\sum_g \mathrm{eq}(r_{\mathrm{hi}}, g)\, d_{\mathrm{blind},g}$ over its 2^{v-c} grid rows is uniform in $\mathbb{F}^L$, the $\mathrm{eq}(r_{\mathrm{hi}}, g)$ summing to 1 over $g \in \{0,1\}^{v-c}$ and the $d_{\mathrm{blind},g}$ being uniform and independent; it is independent of the openings by Lemma 16 and of every support. Then d_q, the sum of this term with terms independent of it, is uniform and independent of the openings and of the ζ_r. □

Lemma 18 (leaves)**.** *Let $S \geq t'+1$ and $\eta\lambda \neq 0$. Given the openings, q, the data of every row and the witness, the part of every table at every unopened column of either tree is uniform on a set of at least $|\mathbb{F}|$ values.*

Proof. Write $x_j = \sum_l a_{j,l} x_{j_l} + \sum_m \psi_{j,m} \bar{x}_m$; for an unopened j, $\psi_j = (\psi_{j,m})_m \neq 0$ since $x_j \notin A$. The symbol of row r at j is $\varsigma_{r,j} + \zeta'_r$ with $\zeta'_r = \langle \psi_j, \zeta_r\rangle$ and $\varsigma_{r,j} = \sum_l a_{j,l} \langle \xi_r, x_{j_l}\rangle + \langle d_r, y_j\rangle$, which the openings and the data determine. The vector $(\zeta_r)_r$ is uniform on $\mathbb{F}^{N(S-t')}$ and the conditioning reveals $\sum_r w_r \zeta_r$ alone; its image $(\zeta'_r)_r$ is then uniform on the affine hyperplane $\sum_r w_r \zeta'_r = \langle \psi_j, \sum_r w_r \zeta_r\rangle$ of $\mathbb{F}^N$. A part is an injective affine image of the ζ'_r of the table's

rows in that tree. In the trace tree those rows number at least two, a data slot and a blind slot, and their ζ'_r range over at least $|\mathbb{F}|$ values of the hyperplane; in the h tree they number at least one, the trace blind row of a table with $v_i = v$, whose weight is nonzero, absorbs the hyperplane constraint, and their ζ'_r range over all of $\mathbb{F}$ each. □

Theorem 19 (hiding)**.** *Let $S \geq t+1$, let every table carry a blind slot, let H be a random oracle, and let the outer argument be δ_{out}-zero-knowledge. The protocol is honest-verifier zero-knowledge with advantage $4TQ_HW/|\mathbb{F}| + 2/|\mathbb{F}| + \delta_{\text{out}}$.*

Proof. The simulator commits the all-zero witness with uniform supports and uniform blinds, forms q by (3) from its own rows, opens the drawn columns, and runs the simulator of the outer argument on the statement these fix: the rows of Theorem 11, with coefficients from the coins and from q. The term $2/|\mathbb{F}|$ excludes $\eta\lambda = 0$. In the honest view replace the pad commitment, the masked scalars and the outer messages by the output of the outer simulator on the honest statement; the views differ by at most δ_{out}, and the witness now enters the view through the pool's part alone, the statement being a function of the coins and q. The opened symbols are uniform and independent of the data in both worlds (Lemma 16). Given them, d_q and the $\langle \xi_q, \bar{x}_m \rangle$ are uniform and the rest of q is the same function of the openings and d_q in both worlds (Lemma 17). The opened parts and leaves are the same function of the opened symbols in both worlds. By Lemma 18, the distinguisher's queries hit the part of some table at some unopened column with probability at most $2TQ_HW/|\mathbb{F}|$ in either world, there being at most $2TW$ such parts; otherwise every digest of such a part is a uniform string in both worlds, and the unopened leaves, the inner nodes and the roots are the same functions of those digests and the opened parts. Conditioned on no hit the two views have one law. □

Appendix B checks Lemmas 14, 15 and 18 by exact ranks over $\mathbb{F}$ on the deployed layout code. The deployed configuration takes $S = t + 1 = 288$, and both parties reject $S \leq t$, at the cost of one element of q. A salt per leaf would also free every leaf, at t salts per opening.

7 Measurements

On ML-KEM-768, ML-DSA and AES the pool changes proof size by −36 to −56%, and on RSA-2048 by −88%. We measured the example workloads of Hekate [16] on an Apple M3 Max, in release builds with the features `parallel`, `blake3` and `table-math`, the verifier with whole-program link-time optimization and one codegen unit; each figure is the best of three runs. Peak memory is the larger of the peak physical footprint and the maximal resident set. The pool runs at $S = t + 1$, the configuration of Table 1; the separate commitments run at $S = t$, to which $S = t + 1$ would add 16 bytes per table.[1] Table 2 collects the zero-knowledge figures.

[1] Hekate 0.36.0 with prover release 0.14.0 for the separate commitments, 0.37.0 with prover release 0.15.0 for the pool; signed public builds `6f09b831` and `4e72a3c1`.

		proof, KiB		verify, ms		prove, ms		peak, MiB
workload	T	separate	pool	separate	pool	separate	pool	pool
ML-KEM-768 sender	9	3 508	1 544	72.1	11.6	567	408	204
ML-KEM-768 receiver	8	3 902	1 884	88.6	13.9	754	552	292
ML-DSA-44	7	2 867	1 547	50.7	10.5	549	432	219
ML-DSA-65	7	2 959	1 568	52.0	10.3	594	472	207
ML-DSA-87	7	3 336	1 812	55.7	11.2	766	618	308
AES-128	3	4 665	2 970	20.1	6.4	1 230	920	770
AES-256	3	4 960	3 198	19.9	6.5	1 330	975	961
RSA-2048	3	14 724	1 810	29.4	8.2	334	204	397
SHA-256, 2^{11} rows	1	540	516	9.4	3.2	69	53	62
SHA-256, 2^{21} rows	1	5 485	5 485	22.7	11.3	10 650	8 720	5 099
Keccak-f, 2^{15} rows	1	858	851	11.9	4.6	195	163	143
Keccak-f, 2^{20} rows	1	3 542	3 544	17.1	7.1	3 730	3 230	2 430
Fibonacci, 2^{20} rows	1	1 146	1 143	6.8	3.7	328	259	252
Fibonacci, 2^{24} rows	1	4 098	4 098	12.9	8.1	5 420	4 060	3 321
Fibonacci, 2^{26} rows	1	8 022	8 021	21.3	12.4	23 320	19 270	13 003

Table 2: Zero-knowledge mode, best of three runs; separate commitments in 0.36.0 at $S = t$, the pool in 0.37.0 at $S = t + 1$.

Every RSA-2048 table has at most 2^{10} rows, and the pool at $c = 15$ packs at least 32 of its columns per slot; no AES table has fewer than 2^{16} rows, and AES gains from the shared fold vector and query set alone. Base mode carries no blinds and no outer argument, and its figures are not tabulated: proof size changes by -37 to -67% on the same seven, and by -90% on RSA-2048. On the seven single-table workloads proof size changes by -0.8 to $+0.1\%$, except SHA-256 at 2^{11} rows, 540 to 516 KiB, while verification time changes by -37 to -66% and prove time by -13 to -25% through changes outside the pool.

8 Remarks

Theorems 11 and 19 take the outer argument's error and simulator as parameters; the deployed instance is a Ligero argument [1] with the rows and parameters of [13]. Outside this note are the per-table ZeroCheck and logUp arguments, whose errors add to $\varepsilon_{\text{pool}}$, and the derivation of the supports, the blinds and the pad, which the prover draws from a seed by a pseudorandom generator and the theorems and the outer instance take as uniform.

Prove time depends on how the prover schedules encoding and hashing, which this note does not describe.

A Necessity of the rules

An interactive simulation over $\mathbb{F}$ with the deployed additive FFT, on four tables of 2^8, 2^7, 2^4 and 2^3 rows at $c = 6$, $S = 16$, $W = 512$ and 24 queries, exhibits the forgery or the inconsistency of each violation in Remark 12. The forging prover that drew η from a copy of the transcript ahead of the last table's claims kept τ_i and passed every later check of the simulation; the verifier of the final build rejected it, at the sumcheck in base mode and at the outer argument's linear test in zero-knowledge mode, because it draws η after those claims. The forging prover that carried a claim error through the eval rounds into the first and the last whole master, where one pair $(1, \lambda)$ would cancel it, was rejected at the line. A forged claim on the 2^3-row table left the target of the constant-padded sumcheck

unchanged. With a short table opened at its own point (r_x, ρ_i), the two points disagreed on one slot.

B Rank computation

Table 3 checks Lemmas 14, 15 and 18, and the hiding of the whole linear view, by exact ranks over $\mathbb{F}$ on the deployed layout code. Every entry of the view and every unopened symbol is a linear form over masking cells, supports and blinds, and witness cells, trace and h; with B the masking part of the view and A its witness part, the view is independent of the witness if and only if $\operatorname{rank}(B) = \operatorname{rank}(B \mid A)$, and a leaf is fixed when the masking part of every symbol it hashes lies in the row space of B. The pools sit at $c = 4$, $W = 32$ and $t = 6$ distinct queries, leaving 26 unopened columns: three tables of 2^6, 2^2 and 2^1 rows, the first on four grid rows and the others packed, two with ring units, two with logUp columns and each with one blind; and one table of 2^2 rows packed on one grid row. The computation first checks the line (4) and the encoding of q against the folded columns.

				fixed leaves	
pool	S	$\operatorname{rank}(B)$	$\operatorname{rank}(B \mid A)$	trace	h
three tables	t	206	206	0/26	26/26
three tables	$t+1$	207	207	0/26	0/26
one table, one grid row	t	34	34	26/26	26/26
one table, one grid row	$t+1$	35	35	0/26	0/26
second table without blind	$t+1$	200	201		
three tables	$t-1$	185	205		

Table 3: Exact ranks over $\mathbb{F}$ of the view at $c = 4$, $W = 32$, $t = 6$. The last two rows are controls with a planted leak.

The linear view hides the witness in all four configurations, and both controls register their leak. At $S = t$ every unopened leaf of the h tree is fixed, and of the trace tree too when one table sits on one grid row; at $S = t + 1$ none is.

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